

- Q.26 Explain the commutator's functions for maintaining the motoring action.
- Q.27 Write about the revolving magnetic field theory.
- Q.28 Discuss about different applications of a stepper motor.
- Q.29 Derive the condition for maximum efficiency in a transformer.
- Q.30 What is starter and explain its types in a DC motor?
- Q.31 What do you understand by armature winding and field winding in a machine?
- Q.32 Write a short note on DC generator.
- Q.33 How to reduce different losses of a transformer?
- Q.34 What are the merits and demerits of a synchronous motor?
- Q.35 What is back electromotive force and its significance?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 How an EMF is induced in a coil when it is rotated in a magnetic field.
- Q.37 Discuss the various characteristics of different types of DC machines.
- Q.38 Derive the relationship between phase current and line current in a Delta configuration of three phase supply.

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3rd Sem / Eltx, EI, IC, Medical Eltx(5th Sem)

Power Eltx, Elect. & Eltx. Engg.

Subject:- Electrical Machines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which type of flux does transformer action need?
- Alternating electric flux
 - Alternating magnetic flux
 - Increasing magnetic flux
 - Constant magnetic flux
- Q.2 What is the main difference between AC and DC servo motors?
- AC servo motors are more efficient than DC servo motors.
 - DC servo motors are more precise than AC servo motors.
 - AC servo motors are more powerful than DC servo motors.
 - DC servo motors are more versatile than AC servo motors.
- Q.3 A refrigerator uses:
- dc series motor
 - Universal motor
 - dc shunt motor
 - Single phase induction motor
- Q.4 What is the working principle of a Transformer?

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- a) Transformer works on the principle of self induction
 - b) Transformer works on the principle of mutual induction
 - c) Transformer works in the principle of ampere law
 - d) Transformer works on the principle of coulomb law
- Q.5 The emf induced in the dc generator armature winding is
- a) AC
 - b) DC
 - c) AC and DC
 - d) None of the above
- Q.6 What is the main component of a servo motor that allows it to control its position?
- a) The motor
 - b) The controller
 - c) The feedback sensor
 - d) The power supply
- Q.7 Induction motor is also known as :
- a) Rotating Transformer
 - b) Singly excited motor
 - c) Asynchronous motor
 - d) All of these
- Q.8 In Dc machine yoke offers
- a) Mechanical protection to the machine
 - b) flux path completion
 - c) produce working flux
 - d) both A and B
- Q.9 Synchronous speed of induction motor depends upon.
- a) Frequency
 - b) No. of stator poles
 - c) Applied voltage
 - d) Both (a) and (b)

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- Q.10 Armature of a DC machine is laminated
- a) To reduce copper losses
 - b) To reduce iron losses
 - c) To improve heat dissipation in armature
 - d) To increase flux density

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define step up transformer.
- Q.12 What is a generator?
- Q.13 Write basic principle of a generator.
- Q.14 What is CVT?
- Q.15 What do you understand by three phase motor?
- Q.16 What is torque angle?
- Q.17 State the use of yoke in a machine.
- Q.18 What is universal motor?
- Q.19 What are the main parts of an induction motor?
- Q.20 Write one application of DC machine.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write down a comparison between a generator and a motor.
- Q.22 What are the main structural differences between single phase and three phase induction motor?
- Q.23 List the factors which are responsible in the speed variation of a DC motor and how?
- Q.24 What is the voltage regulation of a transformer?
- Q.25 Write some advantages of three phase supply over single phase supply.

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